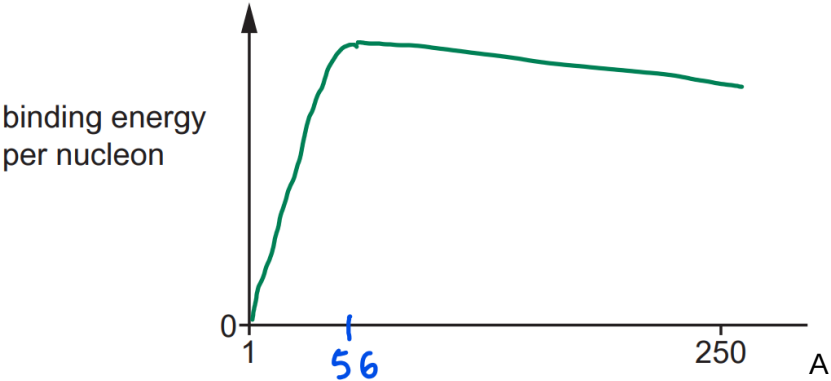
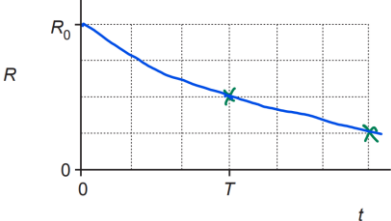


7(a)(i)	Nuclear fusion is the process where two light/small nuclei are combined to produce a heavier/large nucleus with the release of energy.	[1]
(ii)	 binding energy per nucleon 0 1 56 250 A line with a peak at $A \approx 56$ line with steep initial positive gradient on the left of peak and gentler negative gradient at all points to the right of peak and line does not return to 0 binding energy	[1]
(iii)1.	X shown at value of A to the right of the peak	[1]
(iii)2.	Y shown at value of A close to 1	[1]
(iv)	energy from 1 nucleus = $(1.77 \cdot 10^{13}) / (6.02 \cdot 10^{23}) = 2.94 \cdot 10^{-11} \text{ J}$ nucleon number of Z = $93 + 139 + 2 - 1 = 233$ Energy Released = Total Binding Energy Final – Total Binding Energy Initial $2.94 \cdot 10^{-11} \text{ J} = [(1.25 + 1.81) \cdot 10^{-10}] - \text{binding energy of Z}$ binding energy of Z = $[(1.25 + 1.81) \cdot 10^{-10}] - 2.94 \cdot 10^{-11}$ $= 2.77 \cdot 10^{-10} \text{ J}$ binding energy per nucleon $= (2.77 \cdot 10^{-10}) / (233 \cdot 1.60 \cdot 10^{-13}) = 7.43 \text{ MeV}$	[1] [1 – equation to for energy released] [1 – J to MeV]

(b)(i)	$A = \lambda N$ and $\lambda = \ln 2 / T$ Since the number of moles of fluorine-18 is n, the initial number of fluorine-18 nuclei: $N = n \cdot N_A$ $A_0 = \lambda N_0$ 2 photons produced from each decay, so $R_0 = 2 A_0$ $= 2 \lambda N_0$ $= 2 \cdot \lambda \cdot n \cdot N_A$ $= (2 \ln 2) n N_A / T$	[1] [1]
(ii)	 exponential decay curve from $t = 0$ to $t = 2T$, starting at $(0, R_0)$ and with a negative gradient of continuously decreasing magnitude line with negative gradient passing through $(T, R_0 / 2)$ and $(2T, R_0 / 4)$	[1] [1]